

OLAP (Online Analytical Processing)

Introduction

- OLAP is a technology that allows users to perform **multidimensional analysis** of data in data warehouses.
 - It supports complex queries, trend analysis, and decision support.
 - OLAP systems are categorized into **ROLAP, MOLAP, and HOLAP**.
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1. Types of OLAP

a) ROLAP (Relational OLAP)

- Data stored in **relational databases** (tables).
- Uses SQL queries to perform analysis.
- Suitable for very **large data warehouses**.

Advantages:

- Handles large volumes of data.
- Leverages existing relational DBMS.
- Good scalability.

Disadvantages:

- Slower performance due to SQL joins.
 - Limited by relational model constraints.
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b) MOLAP (Multidimensional OLAP)

- Data stored in **multidimensional cubes**.
- Pre-computed aggregations → faster query response.

Advantages:

- Very fast query performance.
- Efficient for complex calculations.
- Intuitive cube representation.

Disadvantages:

- Limited storage capacity (not suitable for huge data).
 - Pre-computation may take time.
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c) HOLAP (Hybrid OLAP)

- Combines **ROLAP + MOLAP**.
- Detailed data stored in relational DB (ROLAP).
- Aggregated data stored in multidimensional cube (MOLAP).

Advantages:

- Balance between storage and speed.
- Can handle large volumes while keeping performance good.

Disadvantages:

- More complex architecture.
 - Higher implementation cost.
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2. OLAP Operations

1. **Roll-up** → Summarize data (e.g., day → month → year).
 2. **Drill-down** → Go into finer details (e.g., year → month → day).
 3. **Slice** → Select a single dimension (e.g., sales in 2023).
 4. **Dice** → Select multiple dimensions (e.g., sales of TVs in 2023 in India).
 5. **Pivot (Rotate)** → Reorient cube for different perspective (e.g., sales by product vs. sales by region).
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3. Advantages of OLAP

- Multidimensional data view.
 - Fast query performance (especially MOLAP).
 - Supports complex calculations and trend analysis.
 - User-friendly for decision-making.
 - Flexible operations (slice, dice, roll-up, drill-down).
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4. Disadvantages of OLAP

- **ROLAP**: Slow for complex queries.
 - **MOLAP**: Storage limitations.
 - **HOLAP**: Expensive and complex.
 - Requires large preprocessing time.
 - Not suitable for real-time transactional data.
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Conclusion

- **ROLAP** → best for large data, scalable but slower.
- **MOLAP** → best for speed, but limited storage.
- **HOLAP** → compromise between both, but complex.
- OLAP systems play a vital role in **data warehousing and decision support** by enabling fast, multidimensional analysis.

